

Tae-Gon Noh

M.S. candidate in Neuroscience

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Summary

M.S. candidate at Seoul National University studying medical AI, sleep physiology, and multimodal biomarkers for neurodegenerative disease progression.

Education

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| M.S. Seoul National University , Neuroscience
candi- • GPA: 3.56/4.0
date • Advisor: Prof. Ki-Young Jung | Seoul, South Korea
Sept 2024 – Aug 2026 |
| B.S. Pusan National University , Biology major, Computer Engineering minor
• GPA: 3.57/4.5
• Mandatory military service, Republic of Korea Army, 2020-2022 | Busan, South Korea
Mar 2018 – Aug 2024 |

Research Experience

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| Seoul National University , Graduate Researcher
Advisor: Prof. Ki-Young Jung.
• Foundation model-based disease progression modeling using brain MRI and PSG.
• Multimodal biomarkers for Parkinson's disease and isolated REM sleep behavior disorder. | Seoul, South Korea
Sept 2024 – present |
| Seoul National University Hospital , Research Intern
Advisor: Prof. Ki-Young Jung.
• Data-driven quantification of sleep and electrophysiological markers in isolated REM sleep behavior disorder. | Seoul, South Korea
Jan 2023 – Aug 2024 |

Research Focus

- Foundation model-based medical AI for disease progression modeling in Parkinson's disease and isolated REM sleep behavior disorder.
- Quantitative sleep and electrophysiology biomarkers from PSG, EEG, EMG, EOG, ECG, brain MRI, and clinical data.
- Survival modeling, heterogeneity modeling, longitudinal analysis, and clinical prediction in prodromal neurodegenerative disease.

Peer-reviewed Papers

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| Clinical, behavioral, and physiological characteristics of isolated rapid eye movement sleep behavior disorder patients with ambulatory dream enactment behaviors
Kim SM, Noh TG , Byun JI, Sunwoo JS, Shin JW, Kim TJ, Jun JS, Schenck CH, Jung KY
10.1016/j.sleep.2026.108968 (Sleep Medicine, 144, 108968) | Jan 2026 |
| Clinical, Physiological, and Behavioral Characteristics of iRBD Patients With Ambulatory Dream Enactment Behaviors
Kim SM, Noh TG , Han SJ, et al
10.1016/j.sleep.2025.107850 (Sleep Medicine, 138, 107850) | Jan 2026 |

Basal electromyographic amplitude in REM and NREM sleep as a predictor of disease progression in isolated REM sleep behavior disorder Noh TG , St. Louis EK, Jung KY, et al 10.1093/sleep/zsaf415 (Sleep, 49(4), zsaf415)	Jan 2026
Diagnostic cutoffs of REM sleep without atonia for identifying isolated REM sleep behavior disorder in a Korean population Noh TG , Byun JI, Hong JK, Yoon IY, Jung KY 10.1016/j.sleep.2026.108811 (Sleep Medicine, 140, 108811)	Jan 2026
Altered functional brain networks in isolated REM sleep behavior disorder during phasic REM sleep Choi KM, Noh TG , Sunwoo JS, Byun JI, Jung KY 10.1016/j.sleep.2025.106724 (Sleep Medicine, 134, 106724)	Jan 2025
Eye blinking characteristics as potential biomarkers in isolated REM sleep behavior disorder during visuospatial attention tasks Kim SM, Choi KM, Noh TG , Kim TJ, Jun JS, Byun JI, Sunwoo JS, Shin JW, et al 10.1016/j.parkreldis.2025.107956 (Parkinsonism & Related Disorders, 138, 107956)	Jan 2025
Continuous REM sleep without atonia quantification improves prediction of phenoconversion to alpha-synucleinopathies in isolated REM sleep behavior disorder Noh TG , Lee S, Lee Y, Kim N, Shin DS, Shin JH, Byun JI, Kim HJ, Jung KY 10.1016/j.sleep.2025.03.013 (Sleep Medicine, 129, 394-401)	Jan 2025
Enhanced delta-gamma phase-amplitude coupling during phasic rapid eye movement sleep in isolated rapid eye movement sleep behavior disorder Noh TG , Choi KM, Jun JS, Shin JW, Byun JI, Sunwoo JS, Jung KY 10.1093/sleep/zsae258 (Sleep, 48(4), zsae258)	Jan 2025
Prediction of phenoconversion into alpha-synucleinopathy in patients with isolated REM sleep behavior disorder using event-related potentials during visuospatial attention tasks Choi KM, Cha KS, Noh TG , Lee S, Shin YW, Byun JI, Jun JS, Shin JH, et al. 10.1093/sleep/zsae308 (Sleep, 48(3), zsae308)	Jan 2025

Conference Papers

Independent Contributions of Aperiodic and Periodic EEG Features as Prognostic Biomarkers for Phenoconversion to Alpha-Synucleinopathies in iRBD Lee EG, Noh TG , Jung KY 10.1093/sleep/zsag091.0799 (Sleep, 49(Suppl 1), A358)	Jan 2026
Resting-State EEG Subtype and Stage Inference Identifies Distinct Prodromal Biotypes and Staging in iRBD Shin JW, Yook SH, Noh TG , Bae S, Jung KY, Kim H 10.1093/sleep/zsag091.0798 (Sleep, 49(Suppl 1), A357-A358)	Jan 2026
Increased phase amplitude coupling during phasic REM sleep in isolated REM sleep behavior disorder Jung KY, Noh TG , Han SJ, et al. onlinelibrary.wiley.com/doi/full/10.1111/jsr.14291 (Journal of Sleep Research, 33(Suppl 1))	Jan 2024
Increased delta-gamma coupling during phasic REM sleep in isolated RBD Noh TG , Lee Y, Kim EY, Sunwoo JS, Jung KY 10.1093/sleep/zsae067.01041 (Sleep, 47(Suppl 1), A447-A448)	Jan 2024

Awards

Paper Award Seoul National University	Dec 2025
Award for Oral Presentation Korean Sleep Research Society	July 2025
Paper Award Seoul National University	Dec 2024
Grand Award for Poster Presentation Korean Society of Sleep Medicine	July 2024
Award for Poster Presentation Korean Sleep Research Society	Nov 2023
Award for Poster Presentation Korean Society of Sleep Medicine	July 2023
2nd Award for CANSAT Korea Korea Advanced Institute of Science and Technology	Aug 2020

Professional Memberships

- American Academy of Sleep Medicine, student member, Dec 2023-present.
- Korean Sleep Research Society, student member, Mar 2023-present.
- Korean Society of Sleep Medicine, student member, Mar 2023-present.
- Korean EEG Study Group, student member, Feb 2023-present.

Skills

Programming: Python, MATLAB, C/C++, Linux

Machine Learning: PyTorch, scikit-learn, variational autoencoders, contrastive VAEs, representation learning, foundation model-based medical AI

Signal Processing and Sleep Analysis: PSG, EEG, EOG, EMG, ECG, audio signal analysis, connectivity analysis, time-frequency analysis

Neuroimaging: Brain MRI preprocessing and analysis, NIfTI processing, atlas-based regional analysis

Genetics and Statistical Genomics: Polygenic risk score construction and analysis, GWAS summary statistics analysis, Mendelian randomization

Biostatistics and Clinical Data Analysis: Regression modeling, survival analysis, longitudinal analysis, clinical prediction modeling